

A Packing of 11 Four-Spaces in $\mathbb{R}^{10}$

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Abstract

We give 11 4-dimensional subspaces of $\mathbb{R}^{10}$ whose smallest squared chordal distance is at least 2.639720. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.617891, obtained by deleting 1 of the 12 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 11 subspaces of dimension 4 in $\mathbb{R}^{10}$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 2.639720 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 10$, $n = 4$. The cell $N = 11$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 2.617891, obtained by deleting 1 of the 12 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(10 - 4)/10 \cdot 11/10 = 2.640000$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m10n4N11.txt`

in <https://github.com/Ge-limin/math719>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 61 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
263972087001650183497945920335546159857237327969581926339688978596954594
197062228324822197317170307865972641547991376061283264888894829934515175
9381340643484890799611996997569491524849918433677
```

denominator:

```
100000000000161053867106452138133633844587578584316055723267065257521539
294916147981039207538365508846466147110399379123381790117867256098081519
9605346137324691657792838734729206896324986591885
```

B Coordinates

11 blocks of 10 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m10n4N11.txt`.

```
564311753761 164484608084 148777964930 -2142610729351912462539 -556517237082 -416652483467 -2182650551222829171190 -277820444835 379305708013 -380647544706
525870266805 465223838970 -407056572610 11697714027318695737752 221802044705 -512176496461 -1661772350436920651132 155945603071 698661073398 309804477635
158498366691 -146930267682 -353154480082 -48297441633275756114 -193702886220 -335618113754 5294321182476323669870 494702248944 60594105065 339307117637
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-354666736073 657969968695 -93494404360 12249232138341946940158 -160802844439 85015583300 -6615100203534082845 276560523686 52491248754 -339388373733
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355100458652 -108682575749 -425090398482 -3124551494513092269114 190555535313 -175741090718 7187268043974151769792 -268840109355 43676291309 388029229679
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```